

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713

COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Jeevan Raaja S (192421005) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Jeevan Raaja S

Annexure A

ANALYTICAL PROBLEM SOLVING

Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
 - ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?
- a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?
- b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

- a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full

notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

1. CRC (Cyclic Redundancy Check)

(i) Determine the transmitted message

Given:

Message = 11100011

Generator Polynomial = $X^3 + 1$

Binary Generator = 1001

Steps:

Append three 0s to the message (degree of polynomial = 3).

11100011 $\rightarrow$ 11100011000

Divide 11100011000 by 1001 using modulo-2 (XOR) division.

The remainder obtained is 100.

Append the remainder to the original message.

Transmitted Message = 11100011100

Explanation:

CRC is an error detection technique used in computer networks. The sender adds a CRC remainder to the original data before transmission. This helps the receiver detect errors that may occur during transmission.

(ii) Error Detection

Steps:

During transmission, assume the leftmost bit changes due to noise.

Received message = 01100011100.

The receiver divides the received data by the same generator (1001).

The remainder is not equal to 000.

Answer:

Since the remainder is non-zero, the receiver concludes that an error has occurred and rejects the received data.

2(a) Throughput Calculation

Given

Bandwidth = 10 Mbps

Frames per minute = 12,000

Bits per frame = 10,000

Step 1

Total bits transmitted per minute

$$= 12,000 \times 10,000$$

$$= 120,000,000 \text{ bits}$$

Step 2

Throughput

$$= 120,000,000 \div 60$$

$$= 2,000,000 \text{ bps}$$

$$= 2 \text{ Mbps}$$

Answer

Throughput = 2 Mbps

Explanation

Throughput is the actual amount of data successfully transmitted over the network in one second. Although the bandwidth is 10 Mbps, the network is effectively transmitting only 2 Mbps.

2(b) Channel Capacity

Given

Bandwidth = 3000 Hz

SNR = 3162

Formula

$$C = B \log$$

$$2^{(1+SNR)}$$

Calculation

$$C = 3000 \times \log_2$$

$$2^{(1+SNR)}$$

$$(3163)$$

$$C \approx 3000 \times 11.63$$

$$C \approx 34,900 \text{ bps}$$

Answer

Channel Capacity $\approx 34.9 \text{ kbps}$

Explanation

According to Shannon's theorem, the maximum data rate depends on both the channel bandwidth and the signal-to-noise ratio. A higher SNR increases the channel capacity.

3. **Hamming (12,8) Error Correction**

Steps

The 8-bit patient data is encoded into a 12-bit Hamming code.

During wireless transmission, one bit changes due to interference.

The receiver checks the parity bits.

The incorrect bit is identified using the syndrome.

The receiver corrects the bit automatically.

Answer

The Hamming (12,8) code detects and corrects a single-bit error without retransmission.

Explanation

This ensures that doctors receive accurate patient information quickly, making the system reliable for real-time patient monitoring.

4. (a) **IPv4 Subnetting**

Given

Network = 172.16.10.0/24

Subnet Mask = /27

Step 1: Number of Subnets

Borrowed bits

$$= 27 - 24 = 3$$

Subnets

$$= 2^3$$

$$= 8$$

Step 2: Usable Hosts

Host bits

$$= 32 - 27 = 5$$

Hosts

$$= 2^5 - 2$$

$$= 30$$

Step 3: Identify Subnet

IP Address = 172.16.10.78

Subnet Range

Network Address = 172.16.10.64

Broadcast Address = 172.16.10.95

Step 4

IP 172.16.10.95 is the broadcast address, so it belongs to the same subnet but cannot be assigned to a host.

Answer

Number of Subnets = 8

Usable Hosts = 30

Network Address = 172.16.10.64

Broadcast Address = 172.16.10.95

172.16.10.95 belongs to the same subnet but is reserved as the broadcast address.

4(b) IPv6 Addressing

Step 1: Compress the Address

2001:0db8:0000:0000:0000:ff00:0042:8329

↓

2001:db8::ff00:42:8329

Step 2: Expand the Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Network Prefix

Using /64

Network Prefix

2001:db8:0:0::/64

Step 4: Host Addresses

Host bits = 64

Possible Hosts

2

64

= 18,446,744,073,709,551,616

Explanation

IPv6 uses 128-bit addresses. A /64 prefix reserves the first 64 bits for the network and the remaining 64 bits for host addresses, allowing an extremely large number of devices.

5. Flow Control

Problem Identified

The main networking problem is Flow Control.

Explanation

The university server transmits question papers at a very high speed (10 Gbps), while many students use slower internet connections (5–100 Mbps). Since the receivers cannot process packets as quickly as they are sent, their buffers become full, causing packet loss, retransmissions, delayed acknowledgments, and increased network congestion.

Solution

Flow control mechanisms, such as the TCP sliding window protocol, regulate the sender's transmission speed according to the receiver's processing capacity. This prevents buffer overflow, reduces packet loss, and ensures reliable data delivery.

Answer

Flow control matches the sender's transmission rate with the receiver's processing speed, improving communication efficiency and reducing network congestion.